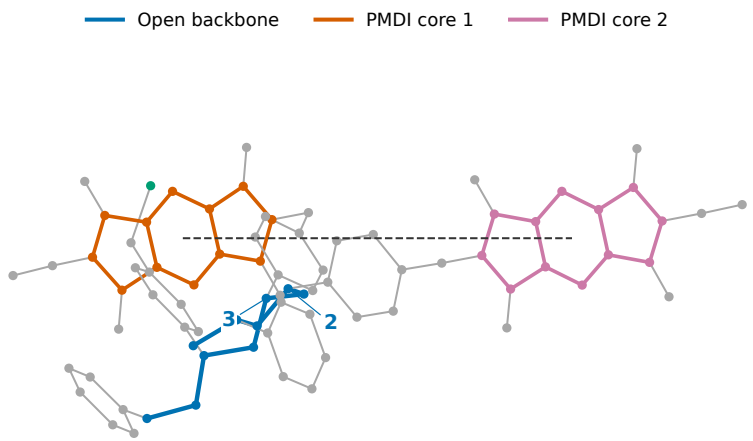
# From redox molecules to grafted polymers

Which molecular advantages survive attachment and nearby redox groups?




3D core-centroid distance: 16.30 Å; plane angle: 69.0°

Actual computed PMDI-grafted open-chain model

**Select a useful material, not just a promising molecule.**

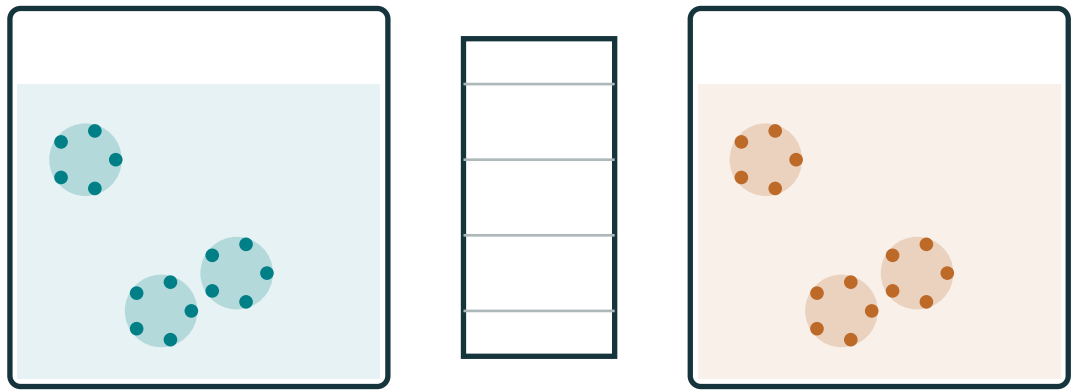
# Larger redox carriers can aid separation

**Store charge on redox motifs.**

**Use a larger carrier to aid separation.**

Route: attach motifs
to a preformed polymer
with benzyl chloride sites.

Conceptual flow-battery architecture



Redox carriers | size-selective separator

**The design spans redox chemistry and polymer context.**

Context: Janoschka et al., Nature 2015, doi:10.1038/nature15746; Montoto et al., JACS 2016, doi:10.1021/jacs.6b06365. Cartoon is illustrative.

# A motif must remain useful after grafting

01

## Useful redox properties

Voltage window
Accessible capacity
Stable charge states

02

## Feasible attachment

Exact precursor
Defined reaction conditions
Core retained after grafting

03

## Weak motif interactions

Spacing and conformation
Motif contact strength
Intersite coupling

**A candidate must pass every required axis.**

# Route screening links UMA starts to solution DFT

## Exact models

Exact graphs
Precursor/product
Oligomer occupancy

→

## Route review

Route evidence
Defined conditions
Screen before QM

→

## UMA

Gas-phase ranking
Seeds to relaxed poses
Charge and spin

→

## DFT (MeCN)

Optimize geometry
Electronic energies
Hessian + qRRHO

## Molecular response

$E_1$, $E_2$ vs $Fc/Fc^+$
Directional solution response

## Grafting $\Delta G$

$\Delta G_{graft}$
Defined base and byproducts

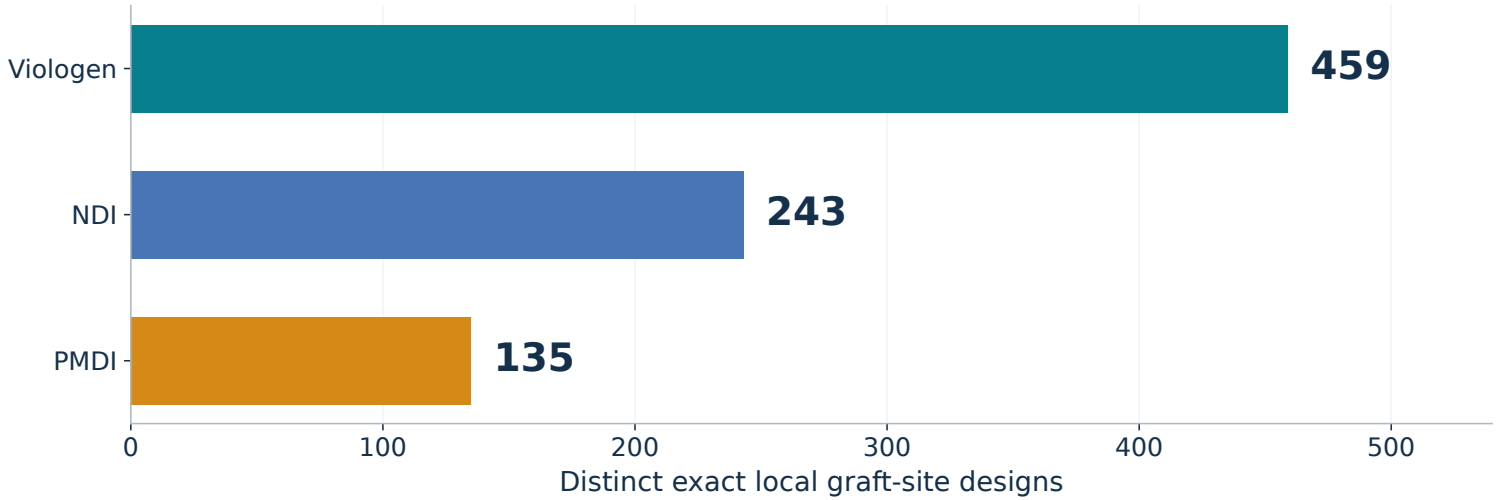
## Neighbor effects

Neighbor $\Delta\Delta G$
Local interaction $E_{int}$

**UMA speeds the search; checked DFT supports the decision.**

UMA is a gas-phase structural accelerator, not the source of the reported solution free energies. Each exact polymer graph is evaluated separately. The tested Nitro force-repeatability gate fails; the standard-inference/V100 profile remains separate.

# A systematic grid anchors a 912-graph pool




**912**

exact graphs in the registry

**Design variables**

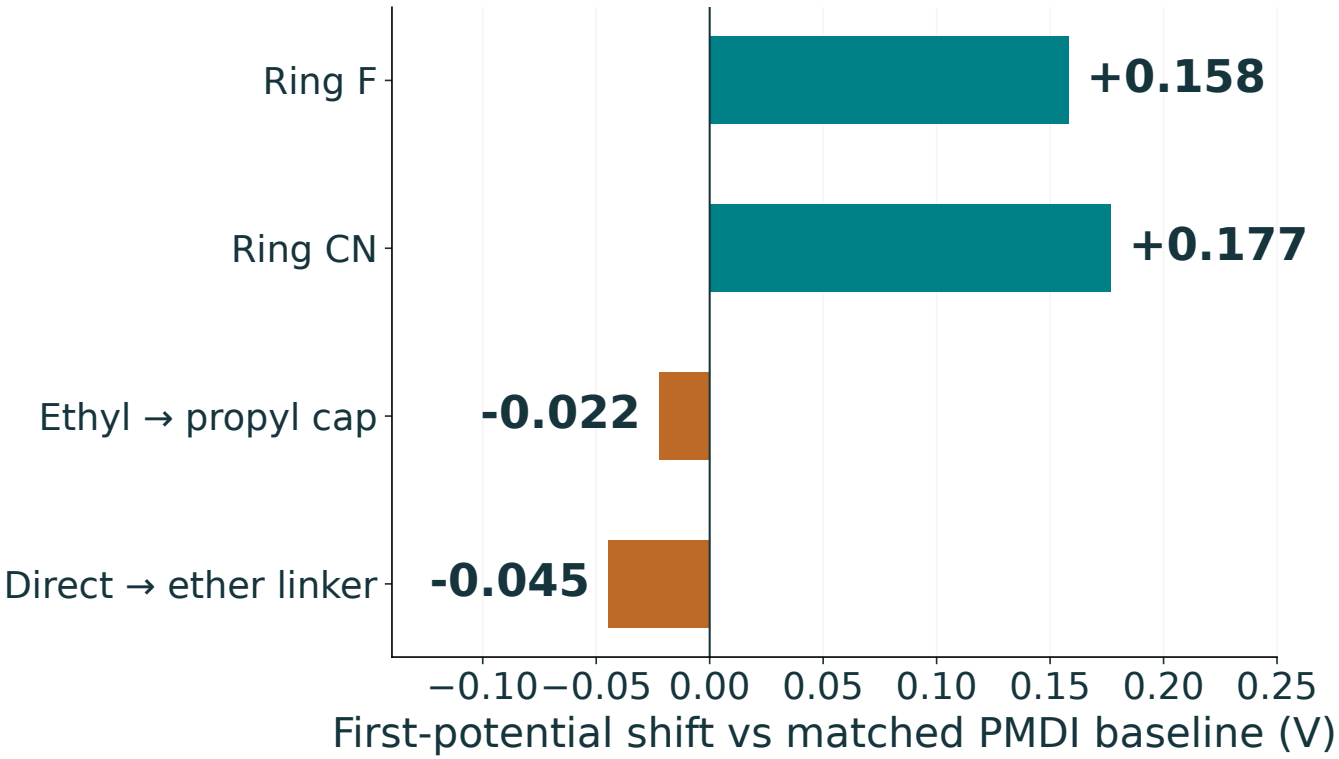
Core
Linker
Cap
Ring substituent

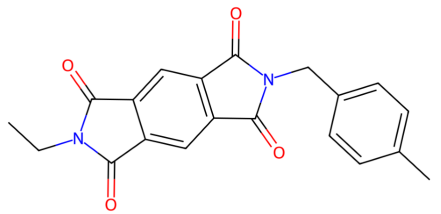
The 837-grid subset spans core, linker, cap and ring choices.

## Enumeration feeds exact-route and quantum checks.

The systematic subset contains 459 viologens, 135 PMDIs and 243 NDIs. The unified registry has 912 graphs and 3 reviewed baseline routes. Eight new direct N-linked phenothiazine F/CN designs remain held and uncomputed. Graph/route QC is separate from quantum or material validation.

# PMDI voltage responds to ring substitution



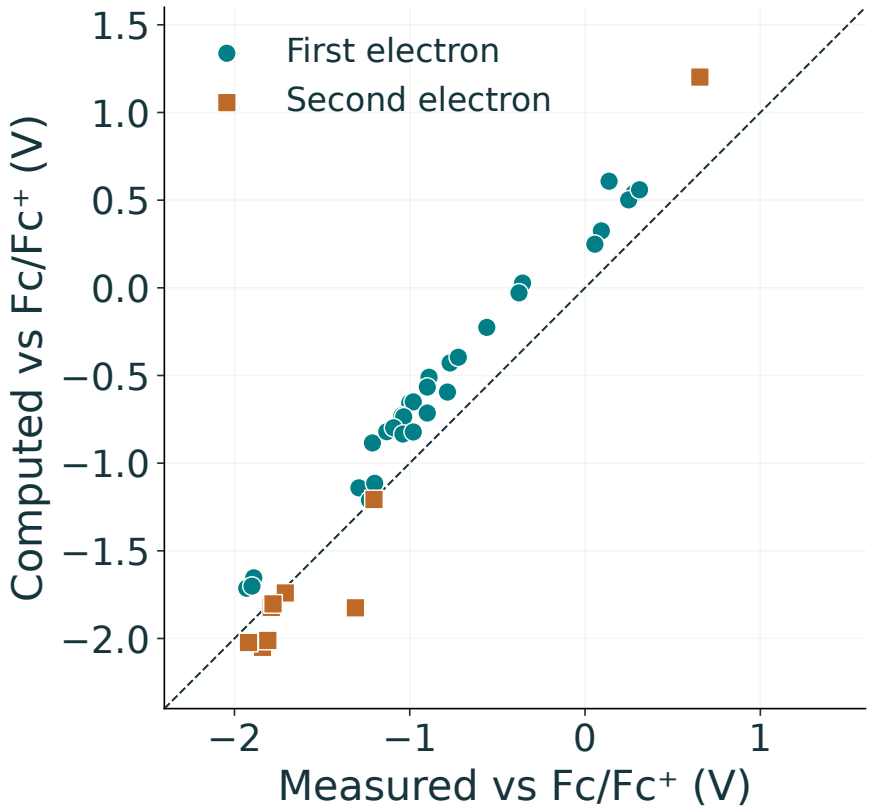

**Same baseline**

PMDI core
Ethyl cap
Local benzyl graft

## Matched comparisons identify molecular tuning levers.

Computed first reductions in MeCN, relative to the exact baseline. Four illustrative contrasts from 18 systematic three-core comparisons; all 26 cohort contrasts reconstruct.

# The benchmark error scale is about 0.25 V




**0.247 V**

Unfitted mean absolute error

**39 endpoints**

**30 molecules**

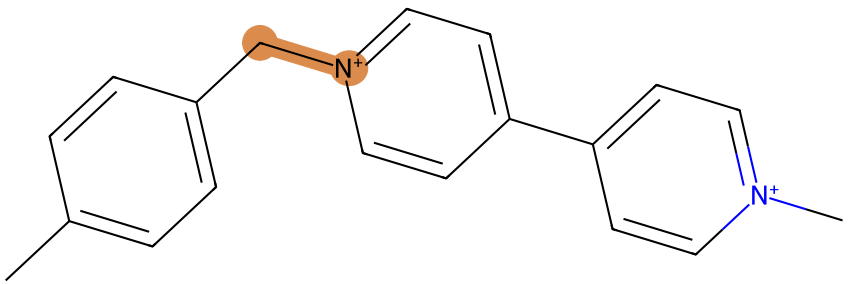
**7 studies**

MeCN; RMSE = 0.279 V

## The benchmark defines a limited molecular accuracy context.

Retrospective exact-free-molecule panel. Molecule-cluster 95% MAE interval: 0.213–0.286 V. No NDI endpoint; grafted-polymer accuracy is not established.

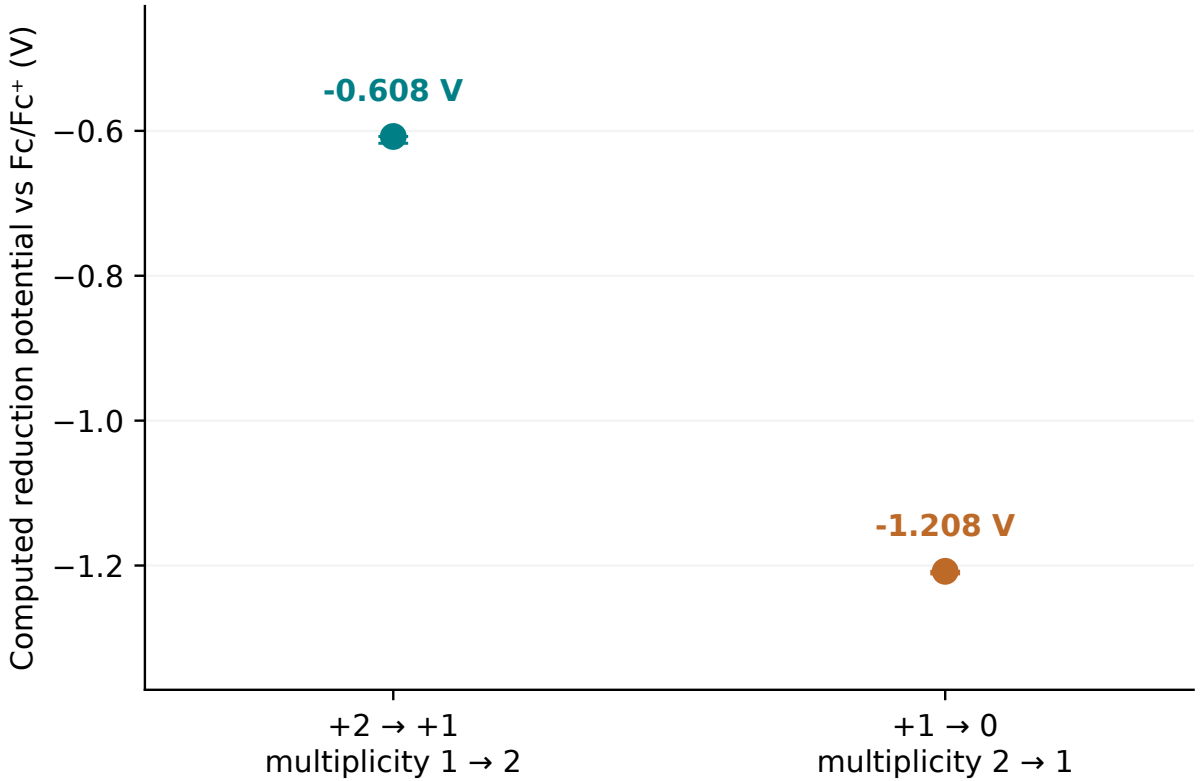
# One exact viologen links redox and grafting




Exact capped graft graph

Grafting $\Delta G = -11.49\,\mathrm{kJ/mol}$

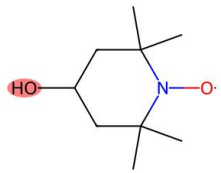
Monomer $\Delta G_{\mathrm{disp}} = +57.91\,\mathrm{kJ/mol}$



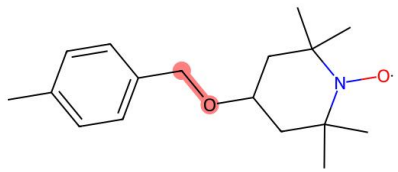

## One graph connects enumeration, route, redox and grafting.

Capped local model, MeCN, 298.15 K; three native states and matched Fc, no fitted offset. Bars show rotor-setting sensitivity. Grafting is a finite local-site separated-ion model; chloride contacts are tested separately; occupied ensembles and coupling remain pending. Drawing shows resting connectivity; no charge/spin localization is inferred.

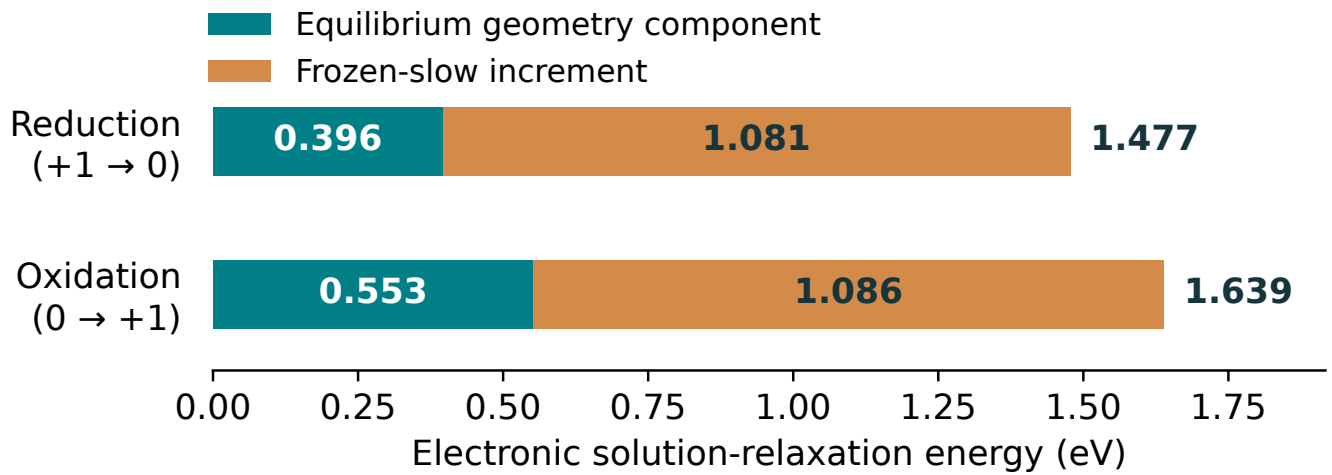
# TEMPO needs a directional response model




4-Hydroxy-TEMPO precursor




4-Methylbenzyl local graft model




Reviewed attachment route

$E_{OX} = +0.579$ V vs Fc

Directional asymmetry: **10.4%**

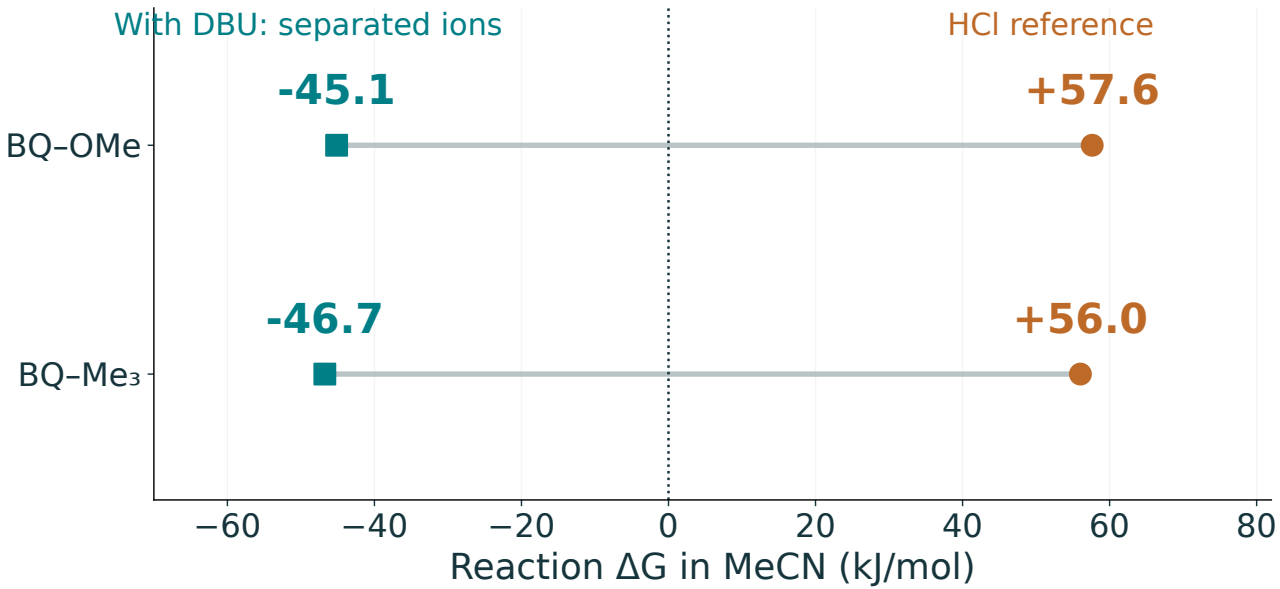
Above the declared 5% diagnostic

**Keep both directions; one symmetric parameter is unqualified.**

Exact capped TEMPO graft; native-SMD frozen-slow electronic response in MeCN at two checked geometry-level minima. Not grafting free energy or a measured rate. No gas-inner or Born-outer term is added.

# The base defines the grafting reaction

$$\text{Motif–OH} + \text{polymer–CH}_2\text{Cl} + \text{B} \longrightarrow \text{polymer–CH}_2\text{–O–motif} + \text{BH}^+ + \text{Cl}^-$$

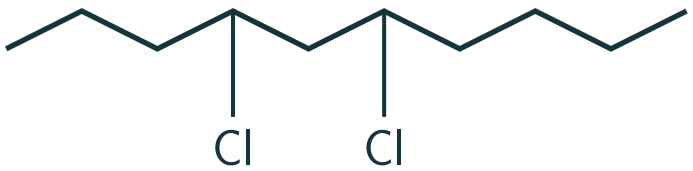



## Base-required BQs remain candidates for further evaluation.

Local graft-site calculations, MeCN, 298.15 K. Computed DBU capture is a shared offset of −102.78 kJ/mol; rankings are unchanged. pKa validation exposes an approximately 23 kJ/mol discrepancy. Signs remain conditional; selectivity and yield are untested.

# Four states isolate the neighbor effect

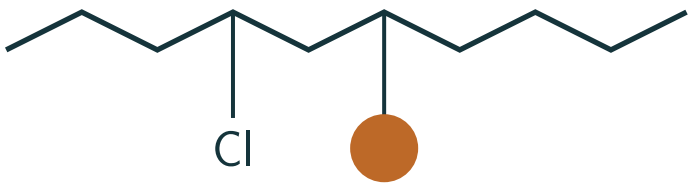
$G_{00}$




$G_{10}$




$G_{01}$




$G_{11}$

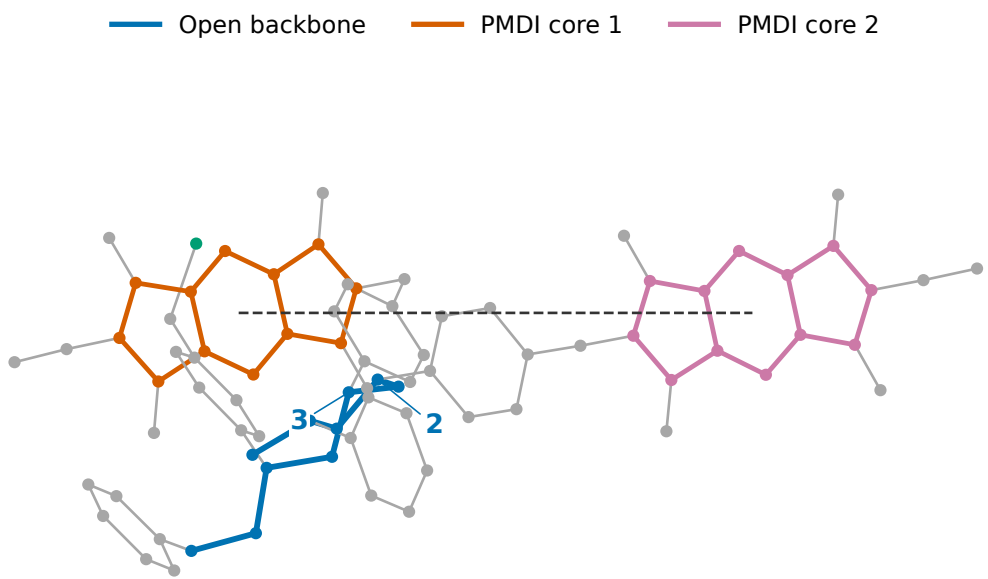


$$\Delta\Delta G = G_{11} + G_{00} - G_{10} - G_{01}$$

**Measure how a nearby graft changes the next grafting step.**

Schematic of the exact occupancy cycle, not an atomistic drawing. All backbones are open chains. Cl denotes an unreacted benzyl-chloride handle; colored circles denote grafts. Actual loading and spacing are unknown.

# A sampled PMDI neighbor has a modest effect




3D core-centroid distance: 16.30 Å; plane angle: 69.0°

**−1.86**

kJ/mol
Neighbor $\Delta\Delta G$

**16.30 Å**

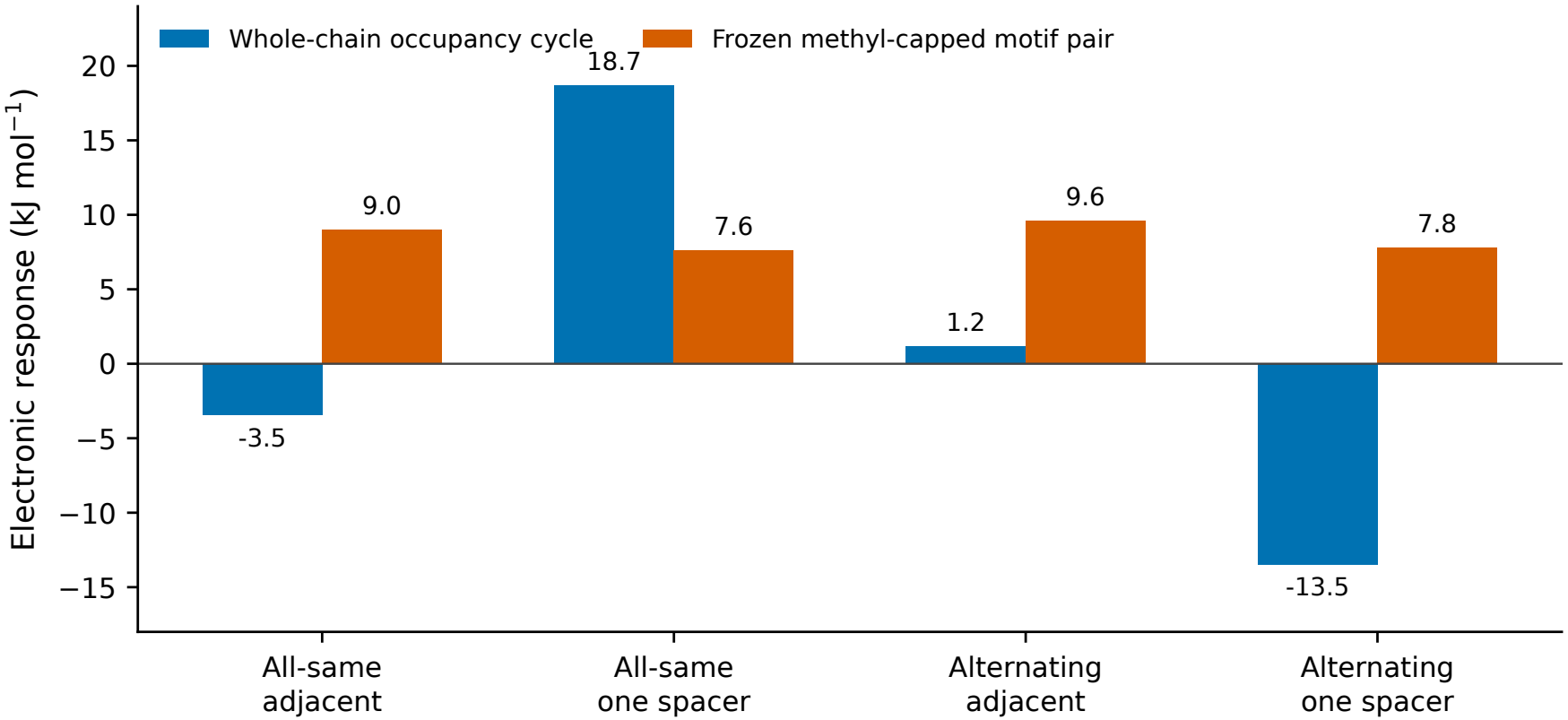
Core separation

Four matched free energies; one minimum per graph.

## The finite PMDI cycle quantifies a polymer-context effect.

Exact ethyl-PMDI design. Local redox: $E_1 = -1.076$, $E_2 = -1.977$ V vs Fc. First-site attachment is 5.34/6.11 kJ/mol less favorable than the capped model. Whole-chain nonadditivity includes backbone and solvent response.

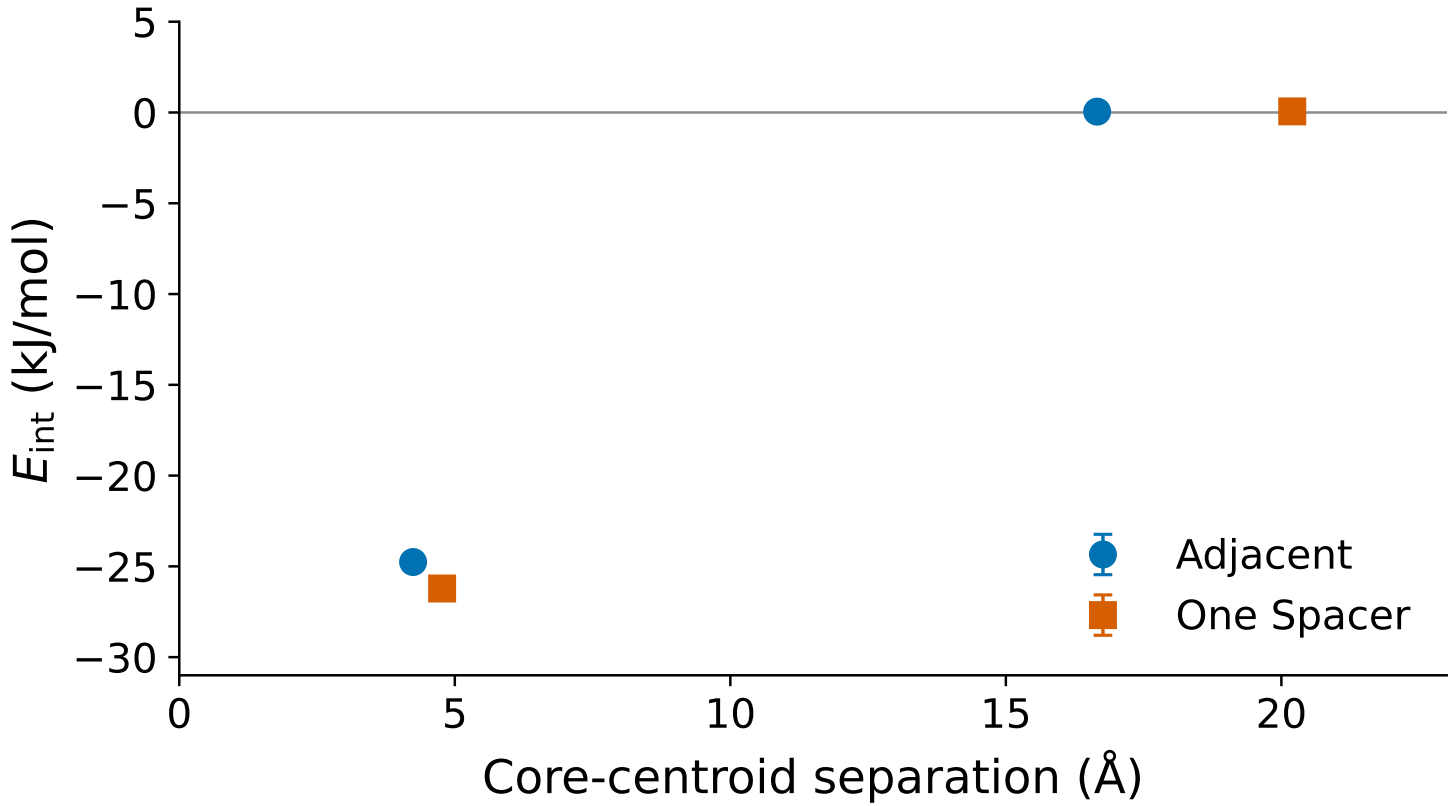
# A favorable chain cycle can coexist with motif repulsion




**The two electronic definitions have opposite signs in two of four frames.**

Twelve A/B/AB triplets and 72 actual SCFs. Frozen methyl caps at 1.07/1.09/1.11 Angstrom; MeCN cap span below 0.003 kJ/mol. Same corrected r$^2$SCAN-D4/SVP smooth-surface model. No binding G, pure-backbone decomposition, coupling or populations.

# A spacer can still permit close contact




**−25 to −26**

kJ/mol
Close phthalimide poses

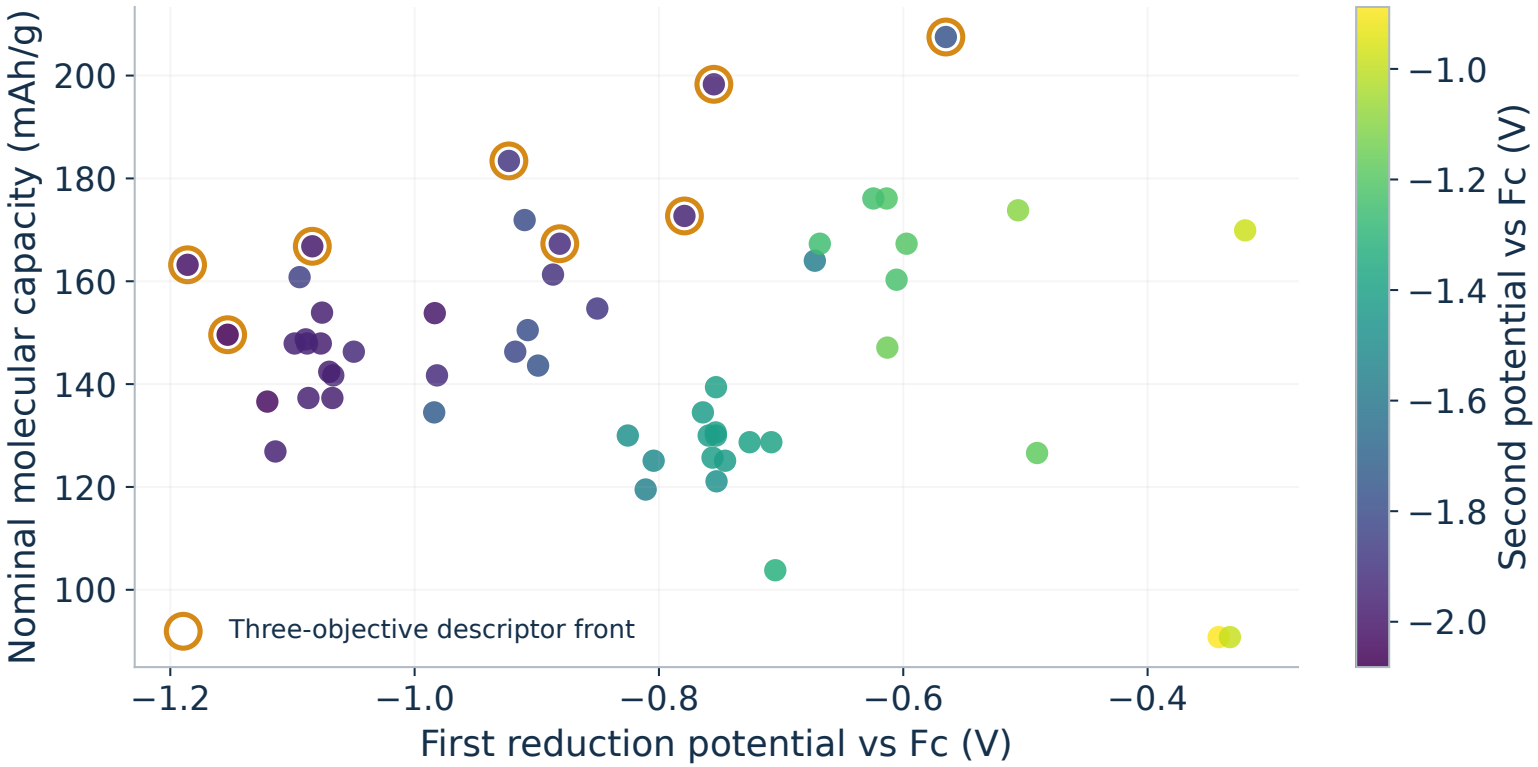
**≈ 0**

kJ/mol
Separated poses

## A sequence spacer does not guarantee weak contact.

Frozen capped-fragment electronic interactions in MeCN, $E_{AB} - E_A - E_B$. Whiskers show cap-length sensitivity, not statistical intervals. Phthalimide is a control. No association free energy, population, basis-superposition correction or electronic coupling is established.

# Every required axis must pass




**8**

Molecular descriptor leaders

**0**

Demonstrated all-axis candidates

## Descriptor leaders still need full material assessment.

Projection of the three-objective comparison over 56 reduction-active local models. Nominal molecular capacity excludes polymer/loading penalties. Route, stability, accessibility, occupancy and interaction evidence remain incomplete.

# Three lessons guide the next designs

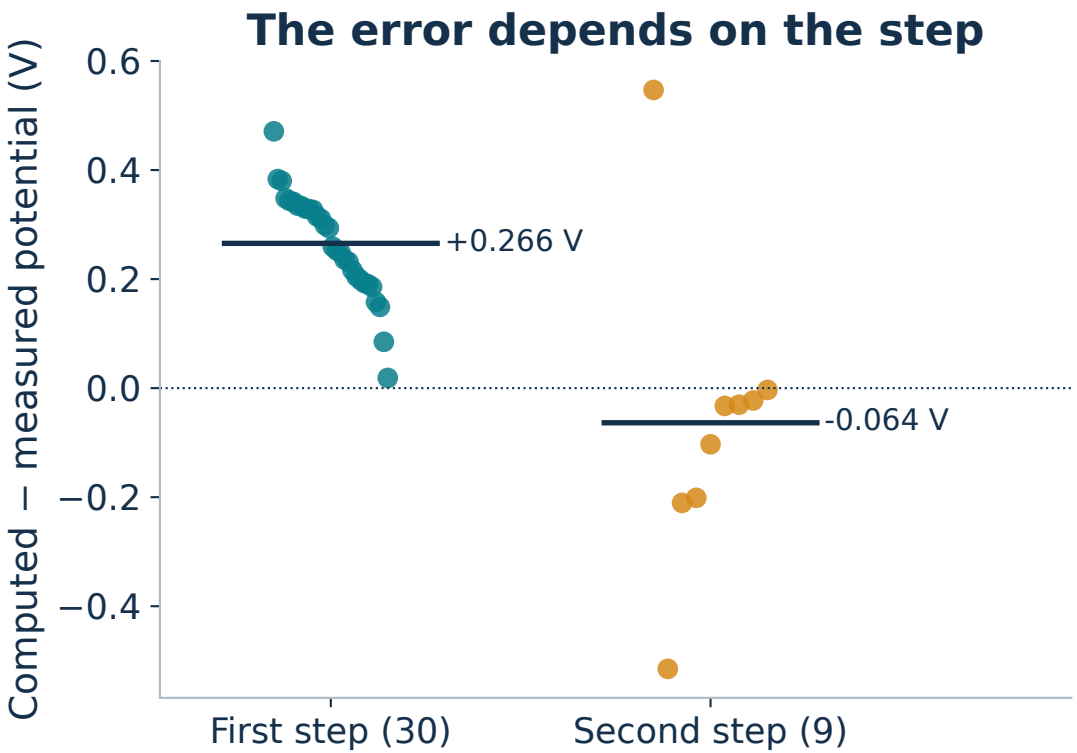
1 **Tune redox chemistry systematically.**
Ring substitutions shift voltage; benchmark error is about 0.25 V.

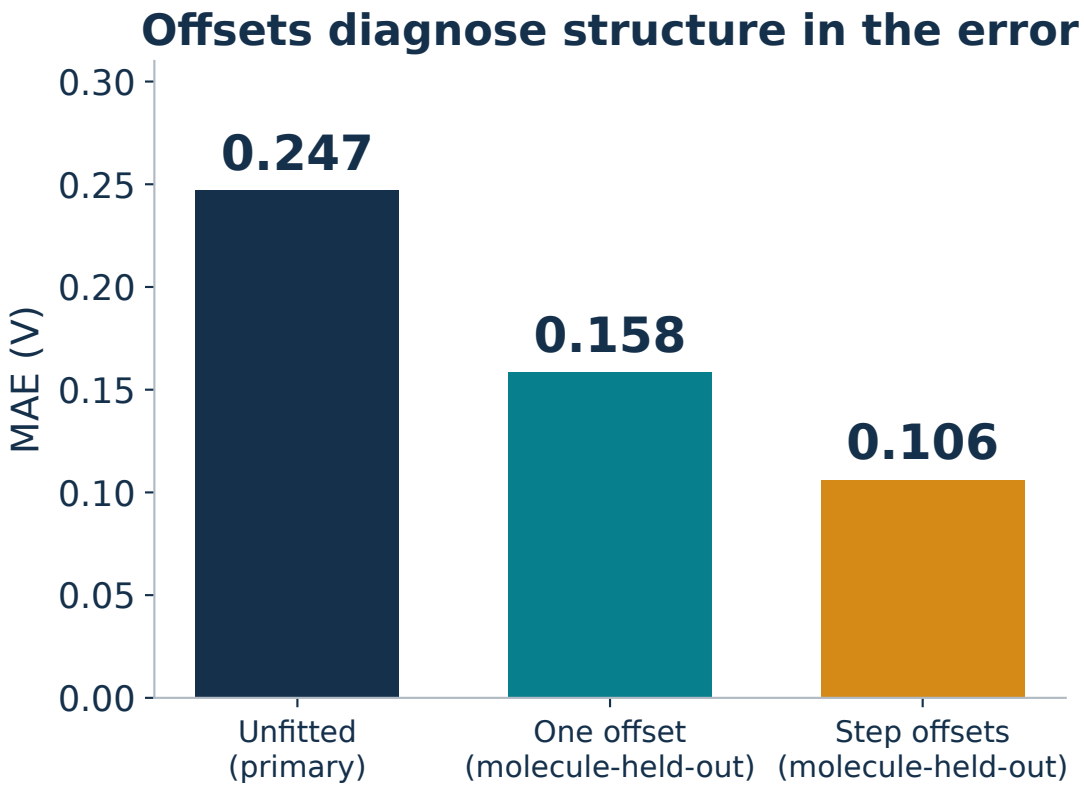
2 **Evaluate a defined attachment reaction.**
Base and speciation affect feasibility; a reference reaction is insufficient.

3 **Distinguish chain response from motif interaction.**
Spacing and conformation change the net chain response.

**Next: broader chemistry and complete polymer-context evidence.**

# Benchmark bias and numerical sensitivity




## Offset diagnostics supplement the unfitted benchmark.

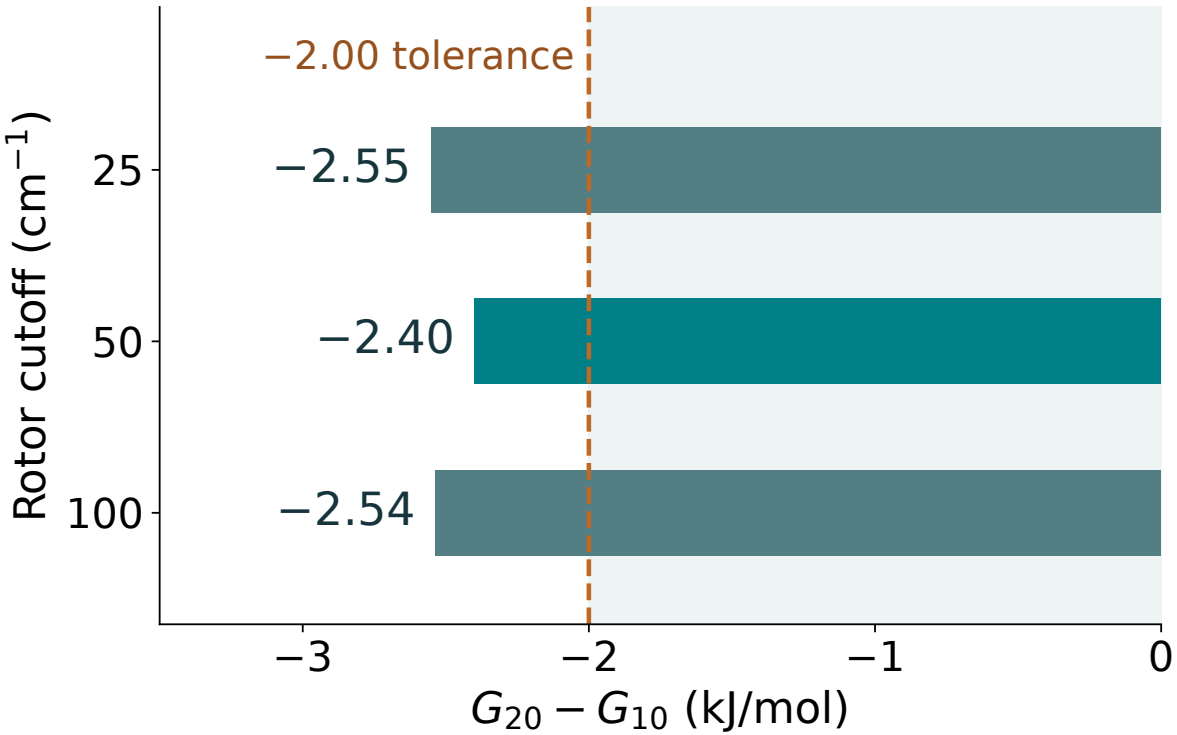
Molecule-held-out MAE: 0.158 V, one offset; 0.106 V, separate step offsets. Five neutral SVPD controls move potentials by −11.7 to −35.1 mV. Neither diagnoses all remaining error sources.

# Computed values and external validation differ

| Quantity | Evidence | Limit |
|---|---|---|
| Redox potential | 39 exact-parent MeCN endpoints | No polymer or NDI accuracy claim |
| Grafting $\Delta G$ | Balanced reactions; experimental base-pKa comparisons | Speciation, selectivity, barriers |
| Polymer neighbor effect | Finite PMDI and viologen cycles | Broader ensembles pending |
| Local motif interaction | 12 triplets / 36 electronic states | No binding G or coupling |

**Keep numerical checks, accuracy and sampling distinct.**

# The first occupied set fails the sampling check




## $10 \rightarrow 20$ starts

Twenty distinct minima
Same result at all three duplicate thresholds

Minimum E lowers 3.88 kJ/mol

Eleven cells incomplete

**Twenty checked starts still fail the declared sampling tolerance.**

First complete corrected phthalimide-control cell: all-same backbone, adjacent sites 2/3; smooth-surface composite MeCN, 298.15 K. Duplicate cutoffs 0.05/0.10/0.20 Angstrom and 0.2 kJ/mol. Source: frozen 31-native-input audit. Balanced occupied cycles and global convergence remain unqualified; no primary-model or measured-population transfer.

# The retained chain context changes the neighbor response

## Four checked finite G cycles

All-same, adjacent: −3.94 kJ/mol

All-same, one spacer: 22.58

Alternating, adjacent: 8.28

Alternating, one spacer: −5.30

Matched atom, charge and reference balance

## What is qualified

Four of four finite ionic cycles

Twelve checked local minima

Own Hessian/nuclear term per state
Occupied ensembles remain pending

**Backbone configuration reverses both finite neighbor responses.**

Corrected $r^2$SCAN-D4/SVP, smooth-surface SMD MeCN, 298.15 K. Each state has its own Hessian/nuclear terms; no explicit spectator ions. No primary molecular-model substitution or material admission. Source: immutable ionic-cycle evidence.

# Explicit chloride contacts test the ion convention

$V^{2+} + Cl^- \rightarrow VCl^+$ Total contact state: $+1$, singlet

| Checked outcome | $\Delta G^\circ_{\mathrm{assoc}}$ (kJ/mol) |
|---|---|
| N0 above, minus mode | 15.61 |
| N0 above, plus mode | 15.42 |
| N1 above | 22.36 |
| N1 below | 22.11 |

**Four outcomes**

**Three duplicate groups**

1 M, MeCN, 298.15 K

Other contacts pending
No population weights

**Finite chloride association is separate from covalent grafting.**

Primary composite MeCN, 298.15 K; each contact has its own native Hessian. N0: benzyl-bearing N; N1: methyl-bearing N. One-atmosphere to 1-M correction: −7.93 kJ/mol, once. N0 outcomes share one group (heavy RMSD 0.00377 Angstrom; energy tolerance 0.2 kJ/mol). Other ions, populations and yield remain unresolved.

# The exact molecular free-energy cohort reproduces

## New numerical evidence

Nine molecular Hessians

81 SCF frames and orbital-spin checks

One analytic chloride partition
Matched electronic and nuclear terms

## Six complete cycles

Potential changes $< 2.45 \times 10^{-8}$ V

Reaction changes $< 1.55 \times 10^{-6}$ kJ/mol

Same source geometries and protocol
Three rotor settings retained

**Numerical reproduction and experimental accuracy remain separate.**

Corrected primary-model exact viologen cohort: two Fc-referenced reductions, three precursor-specific quaternizations and disproportionation. r$^2$SCAN-D4/SVP Hessians plus $\omega$B97M-V/TZVPD electronic terms in SMD MeCN. No polymer-cycle or ensemble admission.

# Reproducibility and model qualification remain explicit

## Available

Code and exact chemical identities

Checkpointed calculations

Frozen evidence and raw-state packages
Relocated arithmetic/Hessian replay

## Remaining

Public data DOI and metadata

Broader deployment portability

Nitro force-repeatability qualification
Complete material-level assessment

**Solution free energies retain the checked DFT protocol.**

Current H100 Nitro repeatability controls fail the stated force threshold; installation is not numerical qualification. V100 uses a separately tested standard-inference fallback.